

# Michael Sobrepera

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## Summary

Problem solver and builder with deep technical expertise in robotics (hardware and software) and AI along with experience in product management and business strategy.

- 7+ years designing, building, testing, and deploying complex robotic systems for human interaction at the University of Pennsylvania and for manufacturing at The Georgia Institute of Technology.
- 3+ years delivering value to large organizations as a management consultant, product manager, and engineer.
- Proven record of leading teams, owning products, developing strategies, and managing complex stakeholder interactions.

Seeking to apply breadth of expertise on a mission-driven team tackling challenging and socially impactful problems.

## Education

### Doctor of Philosophy in Mechanical Engineering

*University of Pennsylvania*

Advisor: Dr. Michelle Johnson

Affiliated with the General Robotics, Automation, Sensing, & Perception Laboratory (GRASP Lab)

Aug 2016 – Apr 2022

*Philadelphia, PA*

### Master of Science in Robotics

*University of Pennsylvania*

Aug 2016 – Aug 2019

*Philadelphia, PA*

### Bachelor of Science in Biomedical Engineering

*Georgia Institute of Technology*

Minor in Computer Science

Aug 2012 – Dec 2015

*Atlanta, GA*

## Experience

### Project Leader

#### Consultant

*Boston Consulting Group (BCG)*

Aug 2024 – present

Apr 2022 – Aug 2024

*Seattle, WA*

- Led 8-person cross-functional team driving upstream product, go-to-market, and stakeholder management for an agentic AI product; accelerated user testing, product iteration, and adoption to maximize value.
- Led 3-person team building BCG's partnership with a hyperscaler to mature a pipeline of AI and digital build opportunities, including acting as the expert to help BCG teams define the capability/product to be built.
- Guided 10+ client organizations to define an AI vision and AI strategy, through workshops with senior leadership, and to develop AI roadmaps and build plans, in coordination with technical leadership, enabling clients to rapidly move toward value capture from AI.
- Built MVP AI products enabling non-technical users to query heterogeneous survey data to improve strategic decision making and analyze incident reports to respond to infrastructure failures.
- Redesigned and implemented the leadership management system and core KPIs of a 15k+ person industrial facility to deliver a turnaround which leadership described as the most impactful shift in a generation.
- Built a large-scale data processing pipeline (in R) to target a multi-million dollar global health investment with a clear focus on need, providing data to a large team of business consultants and client data scientists.
- Developed a strategy perspective in the climate and sustainability space through a combination of desk research and expert interviews which was leveraged widely to support clients on their climate journeys.

### NIH F31 Fellow

#### Graduate Research Assistant

*University of Pennsylvania, Rehabilitation Robotics Laboratory*

Apr 2020 – Apr 2022

Aug 2016 – Apr 2022

*Philadelphia, PA*

- Led end-to-end development (product definition, hardware design and fabrication, software development, and full system integration) for the development of an affordable, socially assistive, upper-humanoid robot to aid in telepresence-based assessment and treatment of patients with upper extremity motor impairments (<https://youtu.be/DDZe1RhcpWY>, <https://github.com/Rehab-Robotics-Lab/FloSystem>).
- Studied patient interactions with telepresence systems which incorporate social robots and impacts on remote assessment and treatment.
- Developed a prototype framework for measuring motor function from video of patients engaging in robot-guided activities using both classical computer vision and machine learning techniques.

**Research Technician II**

Dec 2015 – Jun 2016

**Undergraduate Research Assistant**

Aug 2015 – Dec 2015

*Georgia Institute of Technology, IRIM Technology Transition Laboratory**Atlanta, GA*

- Translated a vision-based 3D object tracker from a research code base to a robust system capable of running reliably, in real-time, and at high resolution to enable robotic manipulation on non-fixed, non-located parts in manufacturing.
- Developed tools for calibrating multiple robot arms to cameras.
- Integrated perception and motion control to track moving targets with a collaborative robot.

**Automation Intern**

May 2015 – Aug 2015

*Eli Lilly and Company**Indianapolis, IN*

- Developed strategy for offline software/hardware/operator-in-the-loop plant simulations (digital twins) for process validation, control code development, control system checkout, operator training, and process improvement.

**Machine Shop Supervisor**

Aug 2014 – May 2015

*Georgia Institute of Technology, TEP Machine Shop**Atlanta, GA*

- Maintained shop, ensuring availability and safety of equipment.
- Guided Master's in Biomedical Innovation and Development students in designing and prototyping of medical devices.
- Supported cardiovascular research labs in development and fabrication of experimental equipment.

**Product Development Engineering Co-Op**

Jan 2014 – Jul 2014

*Unilife Corporation**York/King of Prussia, PA*

- Developed, prototyped, and tested new injectable drug delivery device product concepts.
- Designed and integrated automation equipment for syringe component gluing, assembly, and finish operations.

**Highlighted Skills****Technical Software**

Python, R, Docker, Git, ROS, Computer Vision, System Integration, ML, HRI

**Technical Hardware**

Mechatronics, MCAD, Technical Drawing, Classical+CNC Machining, Additive Manufacturing

**Product & Strategy**

Product Discovery, Roadmapping, Feature Definition, Go-To-Market, Stakeholder Management

**Publications****Peer-reviewed Journal Publications**

- [1] **Michael J. Sobrepera**, Anh T. Nguyen, Ajay Anand, Laura A. Prosser, Sally H. Evans, and Michelle J. Johnson. "Age, Motor Function, and Cognitive Function Influence Preferences for Telerehabilitation Mediated by a Social Robot Augmented with Telepresence". In: *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2025). DOI: [10/pzqz](#).
- [2] **Michael J. Sobrepera**, Anh T. Nguyen, Emily S. Gavin, and Michelle J. Johnson. "Insights into the deployment of a social robot-augmented telepresence robot in an elder care clinic – perspectives from patients and therapists: a pilot study". In: *Robotica* (2024), pp. 1–29. DOI: [10/mpv6](#).
- [3] **Michael J. Sobrepera**, Julie Elfshawy, Anh T. Nguyen, Laura P. Prosser, and Michelle J. Johnson. "Insights on Telecommunication Use by Rehabilitation Therapists Before, During, and Beyond COVID-19". In: *Archives of Rehabilitation Research and Clinical Translation* (2024), p. 100326. ISSN: 2590-1095. DOI: [10/mpv7](#).
- [4] **Michael J. Sobrepera**, Vera G Lee, and Michelle J. Johnson. "The Design of Lil'Flo, a Socially Assistive Robot for Upper Extremity Motor Assessment and Rehabilitation in the Community via Telepresence". In: *Journal of Rehabilitation and Assistive Technologies Engineering* 8 (Apr. 19, 2021), pp. 1–26. ISSN: 2055-6683. DOI: [10/gjrpj8](#).
- [5] **Michael J. Sobrepera**, Vera G Lee, Suveer Garg, Rochelle Mendonca, and Michelle J. Johnson. "Perceived Usefulness of a Social Robot Augmented Telehealth Platform by Therapists in the United States". In: *IEEE Robotics and Automation Letters* 6.2 (2021), pp. 2946–2953. ISSN: 2377-3766. DOI: [10/gh6hkk](#).
- [6] Michelle J. Johnson, **Michael J. Sobrepera**, Enri Kina, and Rochelle Mendonca. "Design of an Affordable Socially Assistive Robot for Remote Health and Function Monitoring and Prognostication". In: *International Journal of Prognostics and Health Management (IJPHM)* 10. Special Issue PHM for Human Health and Performance (2019).

**Peer-reviewed Conference Publications**

- [1] **Michael J. Sobrepera**, Vera G Lee, and Michelle J. Johnson. "Therapists' Opinions on Telehealth, Robotics, and Socially Assistive Robot-Augmented Telepresence Systems for Rehabilitation". In: *2022 9th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*. Seoul, Korea, Republic of: IEEE Press, 2022, pp. 1–8. DOI: [10/mfnn](#).
- [2] **Michael J. Sobrepera**, Vera G. Lee, Suveer Garg, and Michelle J. Johnson. "Feasibility and Acceptability of Remote Neuromotor Rehabilitation Interactions Using Social Robot Augmented Telepresence: A Case Study". In: *2022 International Conference on Rehabilitation Robotics (ICORR)*. 2022, pp. 1–6. DOI: [10/gsgmmk](#).
- [3] **Michael J. Sobrepera**, Enri Kina, and Michelle J. Johnson. "Designing and Evaluating the Face of Lil'Flo: An Affordable Social Rehabilitation Robot". In: *IEEE International Conference on Rehabilitation Robotics*. Toronto, Ontario, Canada, 2019. DOI: [10/ggd87](#).